Monit Sharma

Research Engineer • Singapore Management University • Singapore

Website • Email • LinkedIn • Google Scholar • GitHub

Research & Professional Experience

Research Engineer 2023 – Present
Singapore Management University, School of Computing & Information Systems Singapore

- Lead hybrid quantum, reinforcement-learning, and classical optimization research for vehicle routing and inventory control.
- Collaborated with IBM Quantum on quantum optimization for supply-chain and finance problems.
- Benchmarked standard and qubit-efficient quantum optimization on IBM Heron processors (*Quantum Science and Technology*, 2026); built infrastructure to compare quantum and tensor-network solvers on enterprise-scale workloads.
- Developed AutoQResearch, an LLM-guided system for adaptive policy search in variational quantum optimization (IEEE QCE 2026).

Research & Development Engineer 2022 – 2023
Tata Consultancy Services Mumbai, India

- Solved vehicle-routing instances with 200 customers and 25 vehicles using a quantum-annealing workflow on D-Wave hardware.
- Developed qubit-efficient mappings for mixed-integer linear programming (MILP) supply-chain problems, enabling execution on constrained hardware.

Master's Thesis Researcher 2021 – 2022
Indian Institute of Science Education and Research (IISER) Mohali, India

- Investigated quantum-computing methods for high-energy physics with Dr. Satyajit Jena.
- Demonstrated higher accuracy with single-qubit data re-uploading than comparison algorithms requiring at least three qubits.

Education

Master of Science in Physics; Minor in Data Science Aug. 2020 – May 2022
Indian Institute of Science Education and Research (IISER) Mohali, India

Bachelor of Science in Physics Aug. 2017 – May 2020
Indian Institute of Science Education and Research (IISER) Mohali, India

Research Publications

Journal Articles

Securing the flow: Strategic storage and corridor design for maritime energy security under chokepoint disruptions 2026
With H. C. Lau • *Transportation Research Part E: Logistics and Transportation Review*, published online (12 Oct. 2026) • DOI • arXiv:2605.11990

From Circuits to Hardware: Benchmarking Standard and Qubit-Efficient Quantum Optimization on Real Hardware 2026
With H. C. Lau • *Quantum Science and Technology*, published (Sep. 2026) • DOI • arXiv:2607.11637

Conference Papers

AutoQResearch: LLM-Guided Closed-Loop Policy Search for Adaptive Variational Quantum Optimization 2026
With H. C. Lau • IEEE QCE 2026, Quantum–GenAI Co-Design & Discovery (QGDD) • arXiv:2604.24283

Learning-Based Graph Shrinking for Quantum Optimization of Constrained Combinatorial Problems 2026
With H. C. Lau • QC+AI 2026 (AAAI Workshop on Quantum Computing), Springer CCIS • DOI

Hybrid Learning and Optimization Methods for Solving Capacitated Vehicle Routing Problem 2026
With H. C. Lau • QC+AI 2026 (AAAI Workshop on Quantum Computing), Springer CCIS • DOI • arXiv:2509.15262

Quantum Enhanced Simulation-Based Optimization for Newsvendor Problems 2024
With H. C. Lau, R. Raymond • IEEE QCE 2024 (poster at QIP 2024) • DOI

Quantum Relaxation for Solving Multiple Knapsack Problems 2024
With Y. Jin, H. C. Lau, R. Raymond • IEEE QCE 2024 • DOI

Preprints

Diagnosing Simulation and Hardware Barriers to Cross-Size Transfer in Equivariant Quantum Reinforcement Learning 2026
With H. C. Lau • arXiv:2510.14533

Qubit-Scalable CVRP via Lagrangian Knapsack Decomposition and Noise-Aware Quantum Execution 2026
With H. C. Lau • arXiv:2604.22194

Cutting Slack: Quantum Optimization with Slack-Free Methods for Combinatorial Benchmarks 2025
With H. C. Lau • arXiv:2507.12159

A Comparative Study of Quantum Optimization Techniques for Solving Combinatorial Optimization Benchmark Problems 2025
With H. C. Lau • arXiv:2503.12121

Quantum Monte Carlo Methods for Newsvendor Problem with Multiple Unreliable Suppliers 2024
With H. C. Lau • arXiv:2409.07183

Selected Projects

Peaked Circuit Recovery – Ranked first (score: 2570/2570) on the BlueQubit Peaked Portal leaderboard; recovered 98-bit answers from Quantinuum Helios-1 samples with an auditable, target-blind decoding pipeline.

MettleQ – Developed a quantum-circuit simulator for Apple Silicon GPUs (Metal, MLX), with Qiskit and PennyLane integrations; benchmarked speedups of 1.16× over Aer and 8.37× over Lightning CPU at 20 qubits, with accuracy checks.

Quantum CFD – Built a quantum linear-system solver from primitives; validated on IBM Kingston and Quantinuum Helios-1 and estimated resources for computational fluid dynamics (CFD) matrices.

The Atelier Lab – Developed a local AI-assisted research workbench integrating hybrid retrieval, model routing, and code-change validation to support scientific research and software development.

The Quantum Classroom – Lead workshops and maintain open-source tools and 18 free learning paths in quantum computing, mathematics, and scientific programming, with worked solutions and a browser-based practice lab.

Technical Skills

Languages: Python, C/C++, Java, JavaScript/TypeScript, R, Julia, MATLAB, Mathematica
Quantum Frameworks: Qiskit, PennyLane, Cirq, CUDA-Q, Pytket, TensorFlow Quantum, QuTiP, AWS Braket, D-Wave Ocean, Tensorly-Quantum
Quantum Hardware: IBM Quantum (Heron, Kingston), Quantinuum Helios-1, IonQ, Rigetti, QuEra, IQM, D-Wave quantum annealers
AI & Machine Learning: PyTorch, JAX, TensorFlow, scikit-learn, reinforcement learning, LLM agents & RAG
Scientific Computing & HPC: CUDA, Apple Metal & MLX, MPI, OpenMP, NumPy, pandas
Tools: Git, Docker, HPC clusters

Awards & Presentations

Conference Paper Presentation • *IEEE Quantum Week (QCE), Toronto, Canada* 2026
Research Staff Excellence Award • *Singapore Management University* 2024
Conference Paper Presentation • *IEEE Quantum Week (QCE), Montreal, Canada* 2024
Poster Presentation • *Quantum Information Processing (QIP), Taipei, Taiwan* 2024